*.tmp

Purpose

A `*.tmp` file holds the measured daily temperature time series for a single gauge. One file per station. The filename is listed in `tmp.cli` and the same filename string is matched as the gauge key referenced from `tgage` in `weather-sta.cli`. The reader records daily maximum and minimum air temperature.

Source

- Reader: `src/cli_tmeas.f90`
- Type definition: `src/climate_module.f90`, `type climate_measured_data` (the `tmp` array; `ts` holds daily max, `ts2` holds daily min)

Format

List-directed reads (free format, whitespace-separated). The first two lines are skipped. The third line is the station header. From line four to end of file are the daily records.

Line 1: title

Single string, skipped.

Line 2: column header

Skipped. SWAT+ Editor writes `NBYR  TSTEP  LAT  LONG  ELEV`.

Line 3: station header

Read by `read (108,*,iostat=eof) tmp(i)%nbyr, tmp(i)%tstep, tmp(i)%lat, tmp(i)%long, tmp(i)%elev`.

#	Field	Type	Units	Description
1	nbyr	int	years	Number of years of record in the file.
2	tstep	int	minutes	Time step indicator. Stored but the reader expects daily values regardless.
3	lat	real	degrees	Latitude of the gauge.
4	long	real	degrees	Longitude of the gauge.
5	elev	real	m	Elevation of the gauge.

Records (line 4 onward)

Daily values only:

#	Field	Type	Units	Description
1	year	int	calendar year	Calendar year of the record.
2	jday	int	day-of-year	Julian day (1 to 365 or 366).
3	tmax	real	deg C	Daily maximum air temperature. Stored in `tmp(i)%ts(jday, year_index)`.
4	tmin	real	deg C	Daily minimum air temperature. Stored in `tmp(i)%ts2(jday, year_index)`.

Records are read sequentially. The reader detects the year change after `jday = 365` or `366` and increments its internal year index.

While reading, the routine accumulates monthly sums into `tmp(i)%max_mon` and `tmp(i)%min_mon` (per the `xmon` call) and at end of file divides by the total day count to get the average monthly maximum and minimum. A `-99` flag is treated like any other value at this stage, so missing-data flags carry into the monthly statistics unless the file is clean over the simulation window.

Example

First lines of `refdata/Ames_sub1/ames.tem` (the temperature time series for the Ames reference dataset; the same record layout applies to a `*.tmp` file):

```
AME.Tmp
 NBYR    TSTEP      LAT     LONG     ELEV
   121        0    42.04   -93.89   316
1900    1   -99 -99
1900    2   -99 -99
...
1983    185 32.2    10.6
1983    186 26.7    11.1
```

Station header: 121 years, time step `0`, latitude 42.04, longitude -93.89, elevation 316 m. Early years carry the `-99` missing-data flag for both max and min.

Related files

- `tmp.cli`. Lists each temperature gauge filename.
- `weather-sta.cli`. Binds a temperature gauge to a weather station via `tgage`.
- `file.cio`. Names the `tmp.cli` index and the optional path prefix `in_path_tmp%tmp` applied to every temperature filename.
- `*.tem`. Same record layout, alternate filename extension used by some toolchains.